

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2023

Subject SP5 – Investment and Finance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Attempt all questions beginning your answer to each question on a separate sheet.***
- 4. *Mark allocations are shown in brackets.***
- 5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.***

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) Your company's Chief Investment Officer has asked you to analyse the International Financial Reporting Standards (IFRS).

i) Discuss the motives behind the development of IFRS. (6)

ii) In particular, for institutions trading in investments, outline the key requirements of IFRS 9. (2)

The company you work for is looking to build a framework to monitor market risk and exposure to external economic events such as changes in Government policy.

iii) Describe how you would set up a market risk monitoring and control system in the company. (5)

iv) List the key economic factors by which one may measure the success of a Government's economic policy. (2)

You have been asked to create an investment portfolio with the key underlying principle being the mean-variance portfolio theory.

v) What is the mean-variance portfolio theory? (2)

vi) List the assumptions of mean-variance portfolio theory. (4)

[21]

Q. 2) You have joined an insurance company in the country of Actuarial which has traditionally diversified their investments by geography, splitting their portfolio across domestic and international holdings.

Given the poor returns in the past year given a turmoil in the local equity markets of Actuarial, the Board of the company has suggested that the portfolio needs additional international diversification. However, instead of just focusing on international equities which have been most of the non-local holdings, the Board members have asked for your expert opinion to use international derivatives, such as the futures, options or swap markets.

i) Describe, in turn, how you might make a proposal on achieving international diversification through each of the three alternate types of derivatives mentioned above, with advantages and disadvantages of each type relative to the others. (10)

ii) The mathematical models you have in-house in the company use the normal distribution to model the empirical characteristics of asset prices. However, you know that there are a number of limitations with the normal distribution. In particular, the actual data indicates that there are more peaks and fatter tails in the distribution, increments in price are not always independent and parameters used in the distribution such as market volatility vary over time. Describe how you might go about selecting alternate methods or distributions to model asset prices. (10)

iii) You expect the benchmark Actuarial stock market index to perform poorly in the coming six months. Describe how you can use futures to profit from this, and whether this is better than other alternatives. (4)

[24]

- Q. 3)** A company is attempting to hedge its liabilities for all office rental payments by holding a portfolio of Government bonds until maturity to meet the pre-specified rental payments. For example, the rental payment for each year is matched by the maturity payments from zero coupon bonds maturing at that time.

What are the likely practical problems expected from this arrangement for the company?

[5]

- Q. 4)** During the 1997 Asian financial crisis, Mr. Hoffer, a well-known figure in the hedge fund industry in the past, was struggling of late to find appropriate areas of investment. His investors had started to question his ability to be on top of the speculation ball game. He noticed there was a lot of attention on Asian markets. He eventually decided to sell put options heavily on Thai banks, believing that the Thai Government would not allow their banks to go bankrupt and the fund could pocket the premiums received. His team had been in touch with Thai officials who had indicated that banking default was not a possibility & they would do everything in their power to prevent it. After taking the position, liquidity in the option market gradually waned over time, while their position remained intact, and the Government eventually allowed its banks to default, leading to the folding of Mr. Hoffer's hedge fund.

What were the emotional and cognitive biases exhibited by Mr. Hoffer in this scenario?

[5]

- Q. 5)** Discuss the limitations of resilience testing of an asset liability management strategy.

[5]

- Q. 6)** Mr. A, an individual, is trying to figure out why despite following a perfect asset liability matching strategy at the start of the year, he is struggling to make mortgage payments for his house? He noticed that during the year, his son's education expenses had doubled and they had a vacation abroad that was not planned at the start of the year. Explain in detail which actuarial technique could have been best used to prevent this cashflow shortage?

[5]

Q. 7)

- i) What are "securities transaction tax"? Explain the implications of introducing "securities transaction tax" for all trades in an emerging market.

(4)

- ii) Write a short note on "withholding taxes".

(3)

[7]

Q. 8)

- i) List two arguments each in favor of & against the use of technical analysis as a decision-making tool.

(2)

- ii) When analyzing the patterns for a stock, it was noticed that its 40 day moving average just crossed above its 20 day moving average. What does this indicate about the stock in the context of technical analysis?

(2)

- iii) Discuss two investment approaches that can be created using relative strength analysis. Under what circumstances are each expected to outperform the other?

(3)

- iv) Discuss the primary risk involved in the following scenarios:

- a) A farmer is looking to hedge his wheat produce, due for harvest in 3 months' time, from fall in prices. He short sells a 4 months' wheat futures contract.

- b) An Russian oil producer is looking to hedge fall in price due to be sold in 3 months' time to India & buys put option on BRENT spot since Russian derivatives basket is not liquid.

(2)
[9]

- Q. 9)** The following total return data is provided for a fund portfolio performance and relevant indices:

	Year 1		Year 2		Year 3	
	Index	Fund	Index	Fund	Index	Fund
Index Linked Bonds	6%	9%	8%	3%	2%	6%
Domestic Equities	-5%	3%	6%	5%	17%	16%
Overseas Equities	20%	29%	7%	8%	14%	1%

The fund's strategic benchmark was set at the start of Year 1 as 50% index-linked bonds, 25% domestic equities and 25% overseas equities. It was not amended during the three-year period.

The fund manager adopted a strategy of 25% index-linked bonds, 25% domestic equities and 50% overseas equities at the start of the period and did not re-balance.

- i) Calculate the total returns on the fund and the strategic benchmark over the period. (2)
- ii) Calculate the relative contributions of stock selection and asset allocation at each asset class level as well as total contributions. (6)
- iii) Comment on the results. (2)

[10]

- Q. 10)** The following market data and information has been provided about a pension fund wholly invested in Indian equities:

Date	Market Value of Fund	Index	Dividend Yield of 50
31 Dec 2021	5388	13795	
31 Mar 2022	3614	13372	2.30%
30 Jun 2022	4460	13841	5.20%
30 Sep 2022	5232	14714	0.90%
31 Dec 2022	6149	14243	3.20%

Period	Contribution/Outgo	Investment Income
1	120	95
2	-90	250
3	499	300
4	-129	45

Contributions and investment income all occur on the last day of each quarter.

Calculate for each period and over the full year:

- i)** the time-weighted return. (3)
 - ii)** the money-weighted return. (3)
 - iii)** the index return. (3)
- [9]**
